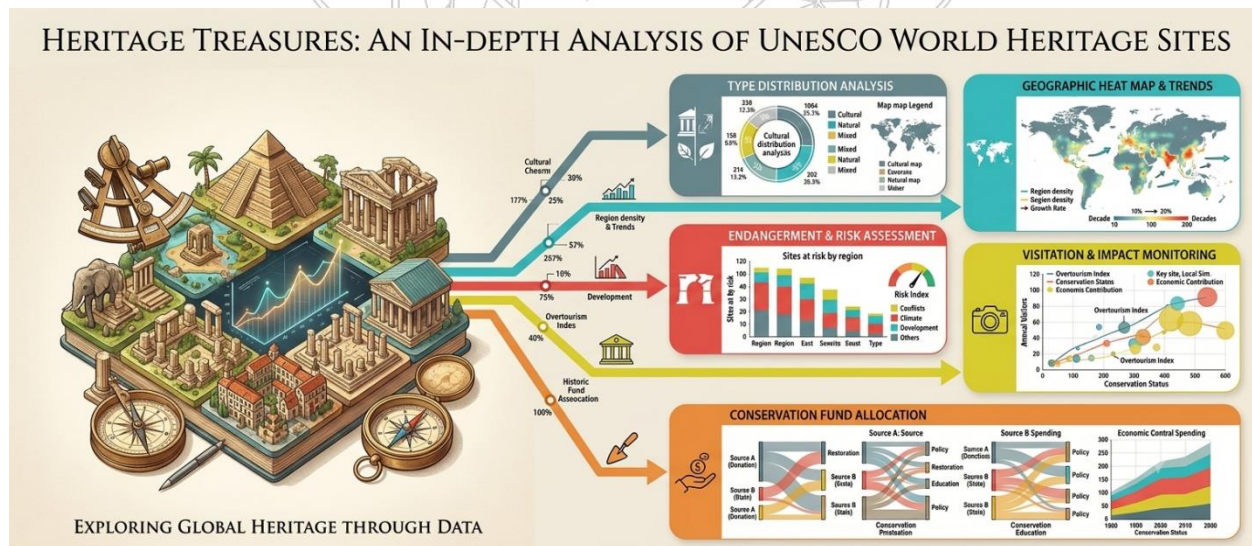


Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

TEAM ID	SWTID-2026-4638
PROJECT TITLE	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
MAX MARKS	3 MARKS

3.3 – Data Flow Diagram



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The data architecture of the "Heritage Treasures" project follows a linear, highly controlled pathway that transforms raw, unorganized archival records into refined, interactive dashboards. The journey begins at the Data Source Layer, which pulls in disparate flat files, geographic coordinate sheets, and historical threat listings from the UNESCO world heritage database. Because these primary inputs contain formatting variations, missing coordinates, and inconsistent country naming conventions, they are passed into an initial pre-processing stage within Tableau's data modeling environment. Here, data cleaning rules handle null values, normalize geographic names, and standardize dates into unified chronological fields. Once cleaned, the records move into the Logical Data Modeling Layer, where physical and logical relationships are defined. Using Tableau's flexible relationships layer, the primary site registry table is linked to the geographic coordinate file and the threat tracking matrix using unique site identification keys. This model preserves the grain of each individual site while allowing for comprehensive, multi-dimensional queries. At this stage, custom analytical calculations—such as regional density metrics and risk index factors—are embedded into the data structure, computing values across various dimensions without altering the underlying raw data records. The final phase is the Visual Presentation and Interaction Layer, which manages how the structured data flows directly into user-facing worksheets and dashboards. When a user interacts with a control—such as adjusting an inscription year slider or clicking on a specific continent—the request is processed by Tableau's high-performance Hyper engine. The engine filters the underlying data model, updates the active calculated fields, and simultaneously refreshes the connected maps, timeline charts, and metric callouts. This responsive data loop ensures that all visual components remain fully synchronized, allowing users to explore complex historical data smoothly and without delay.